

Final Exam — Solution Key

Data Science, AI & Emerging Paradigms

BSIT — DATA SCIENCE

Worked, To-the-Point Answers Grounded in the Course Handouts

Q-1. Differentiate between the three levels of analytics: Descriptive, Predictive, and Prescriptive. (02 marks)

The three levels form a **maturity ladder**: each level answers a harder question and delivers more decision value than the one before it.

Aspect	Descriptive	Predictive	Prescriptive
Question answered	What <i>happened</i> ?	What is <i>likely</i> to happen?	What should we <i>do</i> about it?
Focus	Past & present data	Future outcomes	Best action / decision
Typical methods	Aggregation, summary statistics, dashboards, reports	Regression, classification, forecasting, ML models	Optimization, simulation, recommendation, decision rules
Output	Insight / description	Forecast / probability	Actionable recommendation
Example	“Sales fell 12% last quarter.”	“Sales will fall a further 8% next quarter.”	“Cut price 5% & boost ad spend to recover sales.”

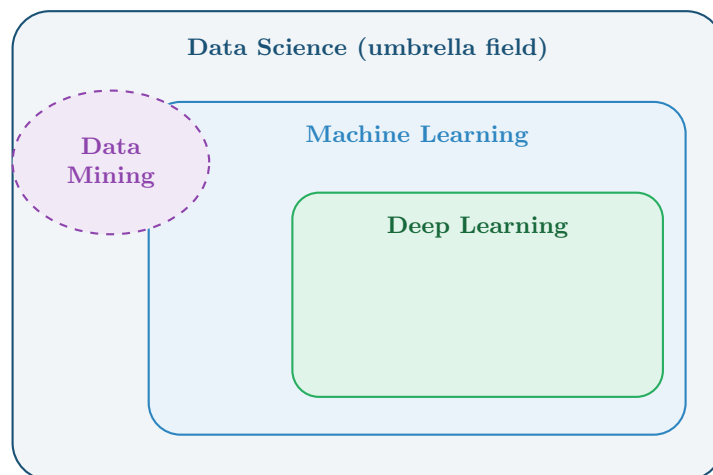
💡 Key Idea

Complexity and business value **increase** from Descriptive → Predictive → Prescriptive. Descriptive looks *backward*, Predictive looks *forward*, and Prescriptive tells you *what to do* — often using the output of the predictive stage as its input.

Q-2. Explain the conceptual differences among Data Mining, Machine Learning, Deep Learning, and Data Science. (02 marks)

These terms are related but sit at different **levels of scope**. Data Science is the broad umbrella; the others are nested capabilities used inside it.

Term	What It Is	Core Focus
Data Science	The broad, multidisciplinary field of extracting knowledge from data	End-to-end lifecycle: collect, clean, analyse, model, visualise, communicate (stats + coding + domain)
Data Mining	The <i>process</i> of discovering patterns / relationships in large datasets	Finding hidden patterns, associations, clusters, and anomalies
Machine Learning	A subset of AI: algorithms that <i>learn</i> from data	Building models that predict or classify without being explicitly programmed
Deep Learning	A subset of ML using multi-layer neural networks	Learning complex patterns & features automatically from images, text, audio



💡 One-Line Summary

Deep Learning $\subset$ **Machine Learning** $\subset$ tools used within **Data Science**; **Data Mining** is the pattern-discovery *process* that overlaps both statistics and ML inside the Data Science workflow.

Q-3. Classify each customer variable as Numerical (Continuous / Discrete) or Categorical (Nominal / Ordinal / Binary). (02 marks)

Variable	Classification	Reason
Customer ID	Categorical — Nominal	Just a label / identifier; the digits carry no magnitude or order (no averaging).
Name	Categorical — Nominal	Text label with no inherent order.
Age	Numerical — Discrete	Counted in whole years (integers). (<i>Continuous if measured exactly.</i>)
Annual Income (\$)	Numerical — Continuous	Can take any value on a scale, including decimals.
Satisfaction Level	Categorical — Ordinal	Ordered categories: Low < Medium < High, but no fixed numeric gap.
Is-Active-Member	Categorical — Binary	Exactly two categories (Yes / No).
Last Purchase Date	Temporal (Interval / Ordinal)	A date: values have a meaningful order and differences (days between), but no true zero.

💡 Watch-Out

A variable stored as *numbers* is not automatically numerical. **Customer ID** looks numeric but is nominal, because arithmetic on it is meaningless. Always ask “do order and arithmetic make sense?” before choosing the type.

Q-4. Compare Single-Agent and Multi-Agent systems. When is multi-agent preferred? Give two advantages, two risks, and a concrete scenario. (06 marks)

Aspect	Single-Agent System	Multi-Agent System
Structure	One agent handles the whole task	Several specialized agents cooperate
Complexity	Simpler to build, monitor, and debug	More modular but more moving parts
Coordination	No inter-agent communication needed	Needs orchestration & message passing
Best for	Focused, well-scoped tasks	Large workflows split into sub-tasks

When is a multi-agent architecture preferred? When the problem can be **decomposed** into distinct sub-tasks that need different skills or tools, when sub-tasks can run **in parallel**, when the workflow is too large for one agent’s context, or when **fault isolation** (one agent failing without taking down the rest) matters.

💡 Two Advantages

- **Specialization & modularity:** each agent is an expert at one job, so it can be improved or replaced independently.
- **Parallelism & scalability:** sub-tasks run at the same time, handling larger workloads faster.

⚠️ Two Risks

- **Coordination overhead:** more communication means more latency, cost, and points of failure.
- **Error propagation & unpredictability:** one agent's mistake can cascade, and emergent behaviour is harder to monitor and control.

📖 Concrete Scenario — an AI Grading & Feedback System for a University

Single-agent design: one agent reads each student's essay, grades it, and writes feedback. Simple, but it does everything sequentially and struggles when the workload (hundreds of scripts) or the rubric grows.

Multi-agent design: an *Orchestrator* splits the work among a *Retrieval agent* (pulls the rubric & reference answers), a *Grading agent* (scores against the rubric), a *Feedback agent* (writes personalised comments), and a *Plagiarism-check agent*.

Advantages realised: the grading and plagiarism agents run in *parallel* across many scripts (scalability), and the rubric logic can be upgraded without touching the feedback writer (specialization).

Risks realised: if the Retrieval agent fetches the *wrong* rubric, every downstream grade is wrong (error propagation), and coordinating four agents adds cost and makes a bad grade harder to trace — so **guardrails, logging, and a human-in-the-loop review** are required.

Q-5. For the store dataset ($N = 8$), compute Support, Confidence, Lift, and Conviction for $\{\text{Laptop, Mouse}\} \rightarrow \{\text{Keyboard}\}$. Why can Confidence mislead, and what does Lift indicate? (06 marks)

Step 0 — The 8 transactions. (✓ = item present in the basket)

Txn	Laptop	Mouse	Keyboard
T1	✓	✓	✓
T2	✓	✓	✓
T3	✓	✓	✓
T4	✓	✓	✓
T5	✓	✓	
T6	✓		✓
T7		✓	✓
T8		✓	✓

Step 1 — Count the baskets we need.

- Contain $\{\text{Laptop, Mouse}\}$ (antecedent A): T1–T5 $\Rightarrow$ **5**
- Contain $\{\text{Laptop, Mouse, Keyboard}\}$ ($A \cup B$): T1–T4 $\Rightarrow$ **4**
- Contain $\{\text{Keyboard}\}$ (consequent B): T1–T4, T6, T7, T8 $\Rightarrow$ **7**

Step 2 — Apply the formulas.

$$\text{Support}(A \Rightarrow B) = \frac{\text{count}(A \cup B)}{N} = \frac{4}{8} = \mathbf{0.50}$$

$$\text{Confidence}(A \Rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)} = \frac{4/8}{5/8} = \frac{4}{5} = \mathbf{0.80}$$

$$\text{Lift}(A \Rightarrow B) = \frac{\text{Confidence}}{\text{Support}(B)} = \frac{0.80}{7/8} = \frac{0.80}{0.875} \approx \mathbf{0.914}$$

$$\text{Conviction}(A \Rightarrow B) = \frac{1 - \text{Support}(B)}{1 - \text{Confidence}} = \frac{1 - 0.875}{1 - 0.80} = \frac{0.125}{0.20} = \mathbf{0.625}$$

Metric	Value	Interpretation
Support	0.50	The rule appears in half of all baskets — fairly frequent.
Confidence	0.80	When Laptop & Mouse are bought, Keyboard appears 80% of the time.
Lift	0.914	< 1: buying Laptop+Mouse makes Keyboard <i>slightly less</i> likely than average — a negative association.
Conviction	0.625	< 1 confirms the rule is weaker than independence would predict.

⚠ Why Confidence Alone Can Mislead

Confidence is a high-looking **0.80**, yet Keyboard already appears in **7 of 8** baskets ($\text{Support}(B) = 0.875$) on its own. So the “80%” mostly reflects that keyboards are bought *anyway*, not a real Laptop+Mouse effect. Confidence ignores the base rate of the consequent, so a common item will *always* look strongly predicted.

💡 What Lift Indicates

Lift corrects for that base rate by comparing the rule to **random chance**:

- **Lift** > 1 → positive association (items bought together more than expected).
- **Lift** = 1.0 → *independence*: the antecedent has *no* effect on the consequent.
- **Lift** < 1 → negative association (as here: 0.914 — they slightly repel).

Here the impressive 0.80 confidence hides a lift below 1: the rule is actually *not* useful.

Q-6. Explain the basic concepts of EDA using the mtcars dataset. (06 marks)

Exploratory Data Analysis (EDA) is the first “get-to-know-your-data” stage: summarise structure, distributions, and relationships *before* modelling. `mtcars` has **32 cars** (rows) and **11 variables** (`mpg`, `cyl`, `disp`, `hp`, `drat`, `wt`, `qsec`, `vs`, `am`, `gear`, `carb`).

1. **Understand the structure.** Check dimensions and types first.

```
dim(mtcars) # 32 rows, 11 columns
str(mtcars) # variable names and types
head(mtcars) # first few rows
```

2. **Summary statistics (central tendency & spread).** Mean, median, quartiles, min/max.

```
summary(mtcars)      # 5-number summary per column
mean(mtcars$mpg)     # ~ 20.09 miles/gallon
sd(mtcars$mpg)       # ~ 6.03 spread
```

3. **Univariate analysis (one variable).** Look at each variable's distribution — shape, skew, spread.

```
hist(mtcars$mpg)     # distribution of fuel economy
boxplot(mtcars$hp)   # spread + outliers in horsepower
```

4. **Categorical / discrete variables.** Use frequency tables for `cyl`, `am`, `gear`.

```
table(mtcars$cyl)    # 4-cyl:11, 6-cyl:7, 8-cyl:14
table(mtcars$am)     # 0 = automatic, 1 = manual
```

5. **Bivariate analysis (relationships).** A scatter plot reveals how two variables move together.

```
plot(mtcars$wt, mtcars$mpg) # heavier cars -> lower mpg
cor(mtcars$wt, mtcars$mpg)  # ~ -0.87 strong negative
```

6. **Correlation matrix.** Rank which variables drive `mpg` most.

```
cor(mtcars)          # mpg vs wt(-0.87), cyl(-0.85), disp, hp
```

7. **Outliers & missing values.** Boxplots flag outliers; `mtcars` has **no** missing values.

```
sum(is.na(mtcars))   # 0 missing values
```

💡 Key Insights from `mtcars` EDA

Average fuel economy ≈ 20 mpg; mpg is **strongly negatively** correlated with weight (`wt`), number of cylinders (`cyl`), and horsepower (`hp`) — i.e. bigger, heavier, more powerful cars drink more fuel. EDA surfaces these patterns *before* any model is trained.

Q-7. RMSE, MAE, and R^2 : give the formula, plain-language meaning, and whether higher/lower is better. Why can RMSE and MAE rank two models differently? (06 marks)

For n observations with actual value y_i , prediction $\hat{y}_i$, and mean $\bar{y}$:

Metric	Formula	What It Measures (plain language)	Better
RMSE	$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$	Typical prediction error in the <i>same units</i> as the target; squaring makes it penalise large errors heavily.	Lower
MAE	$\frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $	The <i>average absolute</i> error; treats every error equally (linearly), so it is robust to outliers.	Lower
R^2	$1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	The <i>proportion of variance</i> in the target explained by the model (1 = perfect, 0 = no better than the mean).	Higher

Why might RMSE and MAE rank two models differently? Because **RMSE squares the errors** before averaging, it punishes a few *large* errors far more than MAE does. So a model with one big miss can look worse on RMSE while looking better on MAE (and vice-versa). Note $RMSE \geq MAE$ always, and the gap grows with the *variance* of the errors (i.e. with outliers).

Worked Example — the Ranking Flips

Five predictions each; look at the absolute errors:

Model A errors = {0, 0, 0, 0, 5} (four perfect, one big miss)

Model B errors = {2, 2, 2, 2, 2} (consistently mediocre)

	MAE	RMSE
Model A	$\frac{0+0+0+0+5}{5} = 1.00$	$\sqrt{\frac{0+0+0+0+25}{5}} = \sqrt{5} \approx 2.24$
Model B	$\frac{2+2+2+2+2}{5} = 2.00$	$\sqrt{\frac{4+4+4+4+4}{5}} = \sqrt{4} = 2.00$

MAE prefers Model A ($1.00 < 2.00$), but **RMSE prefers Model B** ($2.00 < 2.24$). The single large error in Model A is amplified by squaring, so the two metrics *disagree* on which model is best.

Which to Trust?

If a few large errors are especially costly (e.g. badly under-stocking a hospital), prefer **RMSE**. If all errors matter equally and you want robustness to outliers, prefer **MAE**. Report R^2 alongside either to show *how much* of the variation the model actually explains.